INCH-POUND

A–A–59590/25A
20 April 2007
SUPERSEDING
A–A–59590/25
March 7, 2003

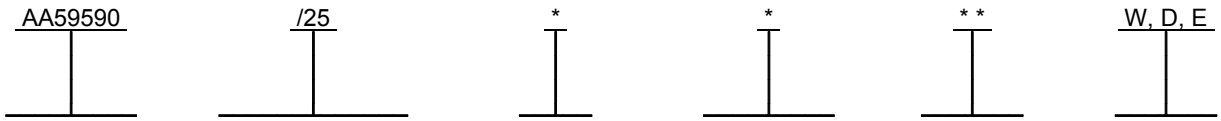
COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

HOLDER, ELECTRICAL CARD, WEDGE RETAINERS, 3 PIECE, SCREW ACTUATED DRIVE, .225 X .260 INCH BODY SIZE, WITH EJECTION ACTION AND VISUAL LOCK INDICATION

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring electrical card holders described herein shall consist of this document and the latest issue in effect of A–A–59590.

CLASSIFICATION/PART IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES).



Example: AA59590/25EJ58 is the PIN for an electroless nickel finished, 5.8 inches (147.3 mm) long card holder with both ejection action and visual lock indication. The card holder contains two counterbore holes for rivet mounting and may be shipped unassembled to allow for rivet installation.

SALIENT CHARACTERISTICS.

Performance. Card holders shall hold the circuit card assembly firmly in its installed position and provide high resistance to loosening due to shock and vibration. Card holders shall also provide maximum thermal transfer from the circuit card assembly to the cold plate.

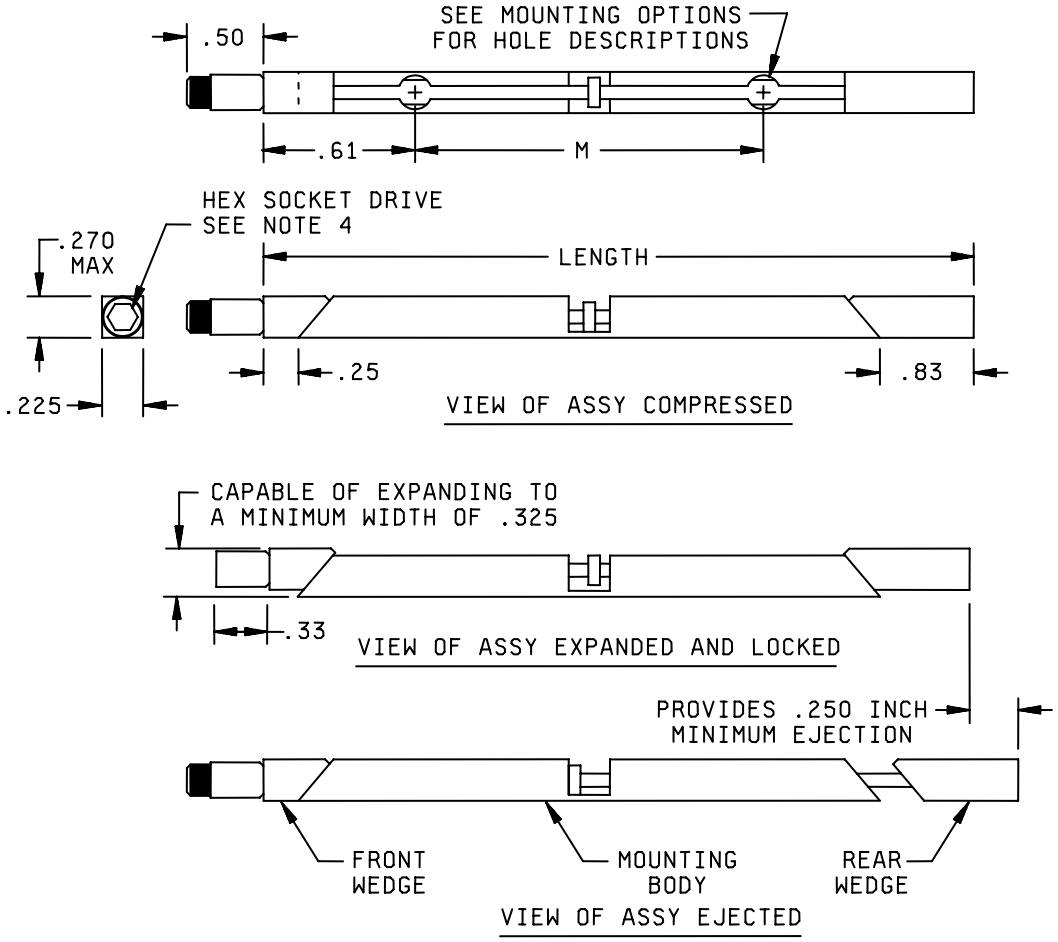
Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified herein (see figures 1, 2, 3, 4, 5, and table I) and meet the general requirements of CID A–A–59590. Card holders shall have an ejection action feature that forces the circuit card assembly to disengage from its installed position. Card holders shall also have a visual indicator to show when the assembly is in its relaxed (unlocked) and expanded (locked) states.

Material. Unless otherwise specified herein, the card holder materials shall be as specified in A–A–59590.

Cold plate dimensions. Recommend cold plate slot width to accommodate the circuit card assembly with attached card holder is .300 inch (7.62 mm) plus the thickness of the printed board of the circuit card assembly (see A–A–59590).

Nominal installation torque. When card holders are used in cold plate applications, the nominal installation torque of each card holder shall be as follows: 6 inch-pounds (0.68 N-m) for assemblies using no options or option “W” and 8 inch-pounds (0.9 N-m) for assemblies using options “D” or “E”.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.225	5.72	.25	6.4	.270	6.86	.33	8.4	.61	15.5
		.250	6.35	.325	8.26	.50	12.7	.83	21.1

NOTES:

1. Dimensions are in inches. Millimeters are given for information only.
2. Unless otherwise specified, tolerances are for ±.02 inch (0.51 mm) for two place decimals and ±.010 inch (0.25 mm) for three place decimals.
3. The tolerance for the mounting hole spacing is ±.005 inches (0.13 mm).
4. The across flats dimension for hex drive socket shall be .094 inch (2.39 mm) for mounting options “J”, “N”, “T”, and “V”. The across flats dimension for hex drive socket shall be a metric dimension of 2.5 mm (.098 inch) across flats for mounting option “G”.

FIGURE 1. Relaxed and expanded dimensions.

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Ejection action (see figure 2). The ejection action shall allow for the initial ejection of the circuit card assembly from its installed position without damaging circuits, components, or the mating connector(s). The minimal amount of ejection distance shall be .25 inch (6.4 mm).

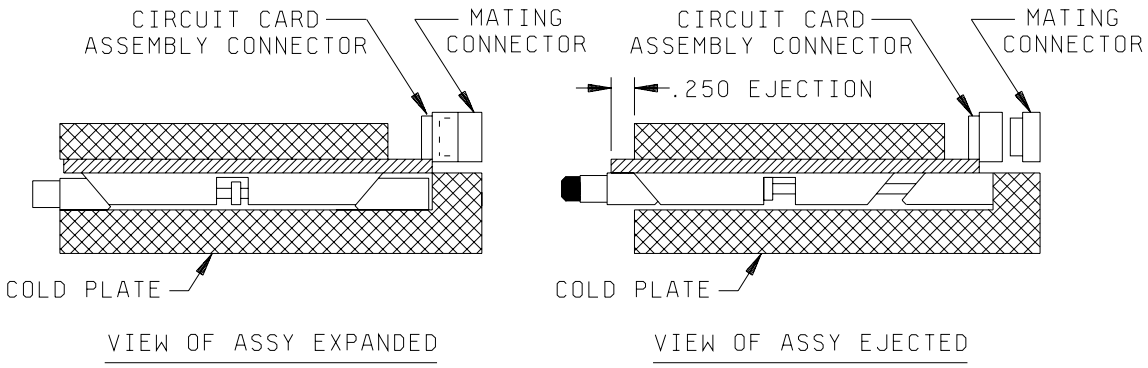


FIGURE 2. View of assembly installed and ejected.

Visual lock indicator (see figure 3). When the card holder is in the relaxed (unlocked) state, the actuating end of the screw shall display a red band on the side of the screw. When the actuating screw on the card holder has been tightened so that the assembly is in the expanded (locked) position, this red band shall be concealed.

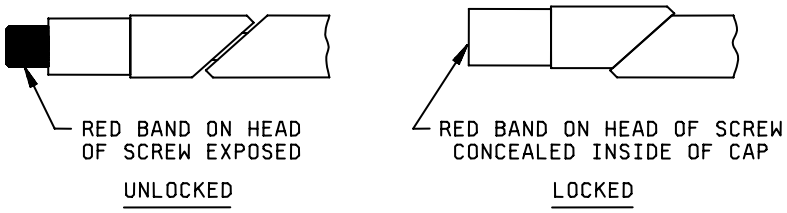


FIGURE 3. Visual lock indicator.

Finish. The finish designator shall be as specified in A–A–59590. The finishes available for this CID specification sheet are as follows: “B” (black anodize), “C” (chemical film), “E” (electroless nickel), or “H” (hard black anodize).

Mounting. The mounting designators shall be as specified in A–A–59590. Except when using mounting option “N”, two mounting holes are required on the card holders. The mounting options available for this CID specification sheet are as follows: “J” (rivet mounting holes), “N” (no mounting holes), “T” (tapped 0-80 UNF 2B holes), “V” (tapped 2-56 UNC 2B holes), or “G” (tapped M2.5 x.45 metric holes). See figure 1 and table I for mounting hole spacing requirements. Card holders using mounting option “J” may be shipped unassembled.

Counterbore holes (for rivet mounting). The holes used for rivet mounting shall be .066/.073 inch (1.68/1.85 mm) diameter, countersunk 100 degrees by .140 inch (3.56 mm) diameter through holes with an access/clearance counterbore of .156 inch (3.96 mm) diameter by .200 inch (5.08 mm) deep.

Length, expanded, and relaxed dimensions. The length designator shall be as specified in A–A–59590 and the lengths available for this CID specification sheet are listed in table I. The length, expanded, and relaxed dimensions shall be as specified on figure 1.

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TABLE I. Assembly dimensions (see figure 1). 1/

PIN length designator	Dimension “LENGTH” ± .02 (0.5 mm)	Dimension “M” ± .005 (0.13 mm)
28	2.8 (71 mm)	1.00 (25.4 mm)
32	3.2 (81 mm)	1.40 (35.6 mm)
38	3.8 (97 mm)	2.00 (50.8 mm)
48	4.8 (122 mm)	3.00 (76.2 mm)
64	6.4 (163 mm)	4.60 (116.8 mm)

1/ Dimensions are in inches. Millimeters, in parenthesis, are given for information only.

Options. Card holders can have options “W” (lockwasher and flat washer), “E” (screw self-locking element), or “D” (combination of options “W” and “E”) as described below. Card holders not requiring these options shall leave the options position in the PIN blank.

Lockwasher and flat washer (see figure 4). A lockwasher and flat washer located under the screw head will provide for additional resistance to loosening from shock and vibration. Card holders requiring a lockwasher and flat washer option shall include a suffix “W” in the PIN (see classification and notes).

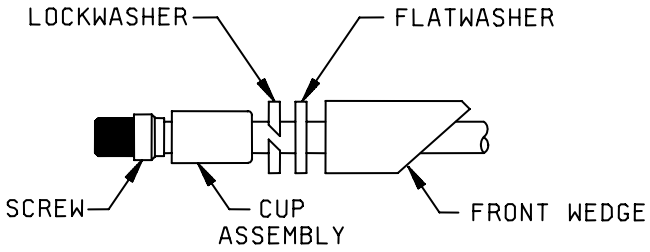


FIGURE 4. Lockwasher and flat washer details.

Screw self-locking element (see figure 5). The use of a screw self-locking element will provide prevailing torque for resistance to loosening from shock and vibration. Card holders requiring a screw self-locking element shall include a suffix “E” in the PIN (see classification and notes).

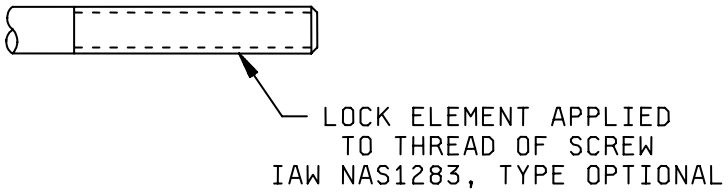


FIGURE 5. Screw self-locking element details.

Lockwasher, flat washer, and screw self-locking element. Card holders requiring a lockwasher, flat washer, and screw self-locking element option shall include a suffix “D” in the PIN (see classification and notes).

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NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

Commercial Item Description

A–A–59590 – Holder, Electrical Card, Wedge Retainers, 3 Piece, Screw Actuated Drive, General Requirements For.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111–5094.)

Other Publications

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

AIA/NAS 1283 – Fasteners, Male Threaded, Self–locking.

(Application for copies should be addressed to the Aerospace Industries Association, 1250 Eye Street, NW, Suite 1200, Washington, DC 20005–3924 or at URL: http://www.aia-aerospace.org.)

Ordering data. Ordering data is as specified in A–A–59590.

Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

Manufacturer CAGE	Manufacturer name and address	Manufacturer contact information
18915	Birtcher Pentair Electronic Packaging 14100 Danielson Street Poway, CA 92064–6898	Telephone: (858) 679–4550 Facsimile: (858) 679–4555 Electronic mail: infobirtcher@pentair-ep.com URL: www.birtcherproducts.com
52094	Calmark Corporation 4915 Walnut Grove Avenue San Gabriel, CA 91776–2099	Telephone: (626) 287–0451 Facsimile: (626) 287–7350 Electronic mail: sales@calmark.com URL: www.calmark.com

Part number supersession data. These CID specification sheet PINs supersedes the following manufacturer's part numbers as shown in table II. The CID PINs listed in table II are only for length designator "28". See table III for CID PIN construction using other available lengths for this specification sheet. This information is being provided to assist in reducing proliferation in the Government inventory system.

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TABLE II. Commercial part number supersession data.

PIN designator AA59590/25	Vendor similar designator or type part number 1/ CAGE 18915	Vendor similar designator or type part number 1/ CAGE 52094
CJ28	45–5073–2.80	VE225–2.80H
CN28	45–5073–2.80–A	VE225–2.80
CT28	45–5073–2.80–T	VE225–2.80T0
CV28	45–5073–2.80–S	VE225–2.80T2
CG28	45–5073–2.80–M	MVE225–2.80TM2.5
CJ28E	45–5073–2.80–L	VE225–2.80HL
CN28E	45–5073–2.80–AL	VE225–2.80L
CT28E	45–5073–2.80–TL	VE225–2.80T0L
CV28E	45–5073–2.80–SL	VE225–2.80T2L
CG28E	45–5073–2.80–ML	MVE225–2.80TM2.5L
CJ28W	45–5074–2.80	WVE225–2.80H
CN28W	45–5074–2.80–A	WVE225–2.80
CT28W	45–5074–2.80–T	WVE225–2.80T0
CV28W	45–5074–2.80–S	WVE225–2.80T2
CG28W	45–5074–2.80–M	MWVE225–2.80TM2.5
CJ28D	45–5074–2.80–L	WVE225–2.80HL
CN28D	45–5074–2.08–AL	WVE225–2.80L
CT28D	45–5074–2.80–TL	WVE225–2.80T0L
CV28D	45–5074–2.80–SL	WVE225–2.80T2L
CG28D	45–5074–2.80–ML	MWV225–2.80TM2.5L
BJ28	45–5073–2.80–B	VEA225–2.80H
BN28	45–5073–2.80–BA	VEA225–2.80
BT28	45–5073–2.80–BT	VEA225–2.80T0
BV28	45–5073–2.80–BS	VEA225–2.80T2
BG28	45–5073–2.80–BM	MVEA225–2.80TM2.5
BJ28E	45–5073–2.80–BL	VEA225–2.80HL
BN28E	45–5073–2.80–BAL	VEA225–2.80L
BT28E	45–5073–2.80–BTL	VEA225–2.80T0L
BV28E	45–5073–2.80–BSL	VEA225–2.80T2L
BG28E	45–5073–2.80–BML	MVEA225–2.80TM2.5L

See footnote at end of table.

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TABLE II. Commercial part number supersession data – Continued.

PIN designator AA59590/25	Vendor similar designator or type part number 1/ CAGE 18915	Vendor similar designator or type part number 1/ CAGE 52094
BJ28W	45–5074–2.80–B	WVEA225–2.80H
BN28W	45–5074–2.80–BA	WVEA225–2.80
BT28W	45–5074–2.80–BT	WVEA225–2.80T0
BV28W	45–5074–2.80–BS	WVEA225–2.80T2
BG28W	45–5074–2.80–BM	MWVEA225–2.80TM2.5
BJ28D	45–5074–2.80–BL	WVEA225–2.80HL
BN28D	45–5074–2.80–BAL	WVEA225–2.80L
BT28D	45–5074–2.80–BTL	WVEA225–2.80T0L
BV28D	45–5074–2.80–BSL	WVEA225–2.80T2L
BG28D	45–5074–2.80–BML	MWVEA225–2.80TM2.5L
HJ28	45–5073–2.80–B3	VEHA225–2.80H
HN28	45–5073–2.80B3A	VEHA225–2.80
HT28	45–5073–2.80B3T	VEHA225–2.80T0
HV28	45–5073–2.80B3S	VEHA225–2.80T2
HG28	45–5073–2.80B3M	MVEHA225–2.80TM2.5
HJ28E	45–5073–2.80B3L	VEHA225–2.80HL
HN28E	45–5073–280B3AL	VEHA225–2.80L
HT28E	45–5073–280B3TL	VEHA225–2.80T0L
HV28E	45–5073–280B3SL	VEHA225–2.80T2L
HG28E	45–5073–280B3ML	MVEHA225–2.80TM2.5L
HJ28W	45–5074–2.80–B3	WVEHA225–2.80H
HN28W	45–5074–2.80B3A	WVEHA225–2.80
HT28W	45–5074–2.80B3T	WVEHA225–2.80T0
HV28W	45–5074–2.80B3S	WVEHA225–2.80T2
HG28W	45–5074–2.80B3M	MWVEHA225–2.80TM2.5
HJ28D	45–5074–2.80B3L	WVEHA225–2.80HL
HN28D	45–5074–280B3AL	WVEHA225–2.80L
HT28D	45–5074–280B3TL	WVEHA225–2.80T0L
HV28D	45–5074–280B3SL	WVEHA225–2.80T2L
HG28D	45–5074–280B3ML	MWVEHA225–2.80TM2.5L

See footnote at end of table.

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TABLE II. Commercial part number supersession data – Continued.

PIN designator AA59590/25	Vendor similar designator or type part number 1/ CAGE 18915	Vendor similar designator or type part number 1/ CAGE 52094
EJ28	45–5073–2.80–EN	VEEN225–2.80H
EN28	45–5073–2.80ENA	VEEN225–2.80
ET28	45–5073–2.80ENT	VEEN225–2.80T0
EV28	45–5073–2.80ENS	VEEN225–2.80T2
EG28	45–5073–2.80ENM	MVEEN225–2.80TM2.5
EJ28E	45–5073–2.80ENL	VEEN225–2.80HL
EN28E	45–5073–280ENAL	VEEN225–2.80L
ET28E	45–5073–280ENTL	VEEN225–2.80T0L
EV28E	45–5073–280ENSL	VEEN225–2.80T2L
EG28E	45–5073–280ENML	MVEEN225–2.80TM2.5L
EJ28W	45–5074–2.80–EN	WVEEN225–2.80H
EN28W	45–5074–2.80ENA	WVEEN225–2.80
ET28W	45–5074–2.80ENT	WVEEN225–2.80T0
EV28W	45–5074–2.80ENS	WVEEN225–2.80T2
EG28W	45–5074–2.80ENM	MWVEEN225–2.80TM2.5
EJ28D	45–5074–2.80ENL	WVEEN225–2.80HL
EN28D	45–5074–280ENAL	WVEEN225–2.80L
ET28D	45–5074–280ENTL	WVEEN225–2.80T0L
EV28D	45–5074–280ENSL	WVEEN225–2.80T2L
EG28D	45–5074–280ENML	MWVEEN225–2.80TM2.5L

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.

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TABLE III. Example of PIN with available length designators.

PIN designator AA59590/25	Vendor similar designator or type part number 1/ 2/ CAGE 18915	PIN designator AA59590/25	Vendor similar designator or type part number 1/ 2/ CAGE 52094
EJ28	45–5073–2.80–EN	EJ28	VEEN225–2.80H
EJ32	45–5073–3.20–EN	EJ32	VEEN225–3.20H
EJ38	45–5073–3.80–EN	EJ38	VEEN225–3.80H
EJ48	45–5073–4.80–EN	EJ48	VEEN225–4.80H
EJ64	45–5073–6.40–EN	EJ64	VEEN225–6.40H

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.

2/ Other lengths are available on request.

MILITARY INTERESTS:

Custodians:
Army – CR
Navy – EC
Air Force – 11
DLA – CC

Review activity:
Air Force – 99

CIVIL AGENCY COORDINATING ACTIVITY:

GSA – FSS

Preparing Activity:
DLA – CC

Project 5998–2007–004

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